A-A-1012
March 17, 1983

COMMERCIAL ITEM DESCRIPTION

CASTER, SWIVEL (INDUSTRIAL DUTY)

The General Services Administration has authorized the use of this commercial item description in preference to type II, group B, style 4A of Federal Specification FF-C-88.

This description covers industrial quality swivel casters suitable for intermitten operation on nonpowered material transport equipment.

Salient characteristics:

Casters furnished under this commercial item description shall have flat mounting plates and ball-bearing swivels. They shall be of the size and height specified and shall conform to the dimensions and load rating defined by figure I and table I.

Size A - 2-1/2 inches diameter wheel
Size B - 3-1/2 inches diameter wheel

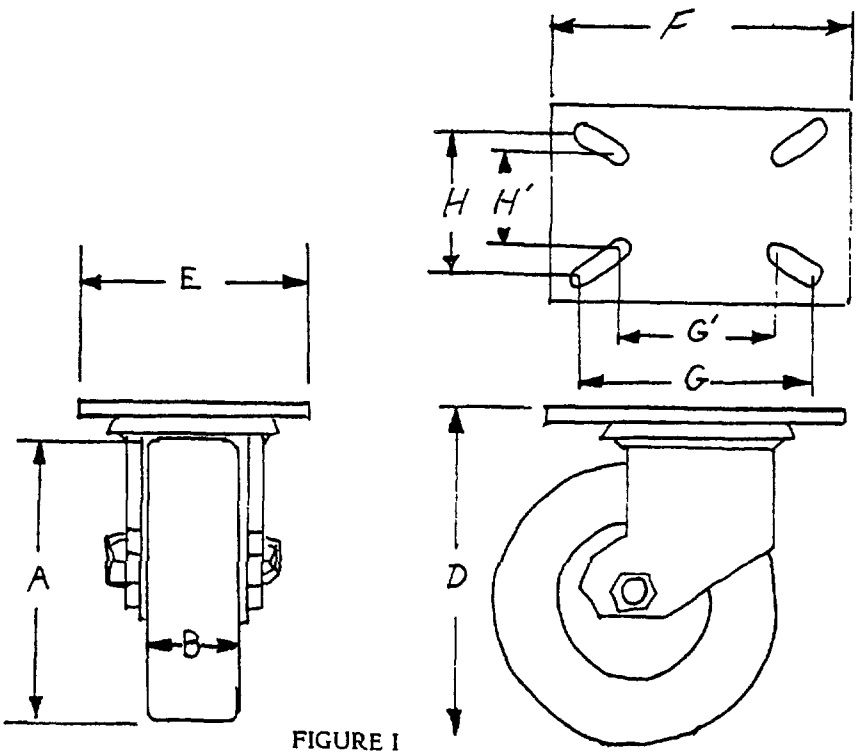


FIGURE I

TABLE I

Size	Load	B (min)	D ± 3/32	E ± 1/8	F ± 1/8	G ± 1/32	H ± 1/32
A	200	1-1/8	3-5/16	2-1/2	3-5/8	2-1/8	1-3/4
		1-1/4	3-3/4	3-1/2	3-5/8	2-1/8	1-3/4
B	250	1-1/4	4-5/8	3-3/4	4-1/2	3 to 3-7/16*	2-3/8 to 3*
			4-19/32	2-1/2	5	4	1-3/16
			4-3/4	2-1/2	4-1/2	2-7/8 to 3*	1-3/4 to 2-3/8*
			4-1/2	2-1/2	3-5/8	2-7/8	1-3/4

*Slotted

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Swivel bearings. The swivel mechanism of the casters shall have hardened, but not work hardened bearings which are polished and plated to resist corrosion.

Wheels. Casters shall have wheels which are of high quality hard rubber that is nonmarking, high impact and oil resistant. The minimum hardness shall be 80 ± 5 points on the D scale, Shore durometer in accordance with ASTM D 2240.

Wheel bearings. Casters shall be provided with oil impregnated wheel bearings which are formed of powdered metal with a high bronze content. The bearings shall be fixed to the full width of the wheel hub. When roller bearings are specified, the outer raceway or sleeve shall be of hardened steel.

Axle. The assembly shall be a steel hex bolt and nut. The minimum bolt diameter shall be 5/16 inch for size A casters and 3/8 inch for size B casters. The axle shall be cold formed and shall be zinc or cadmium plated. During operation, the axle shall not rotate, loosen or bind the wheel.

Spanner bushing. The bushing shall be of hardened steel and shall be ground or polished to form a smooth inner race for the bearing. Minimum outside diameter - 9/16 inch, minimum wall thickness - 0.072 inch.

Thread guards. When thread guards are specified, the minimum wall thickness shall be 1/32 inch.

Swivel assembly. The ball bearing assembly shall be single or bi-level with a smooth nonbinding operation. The raceways shall be smooth and hardened and shall have a minimum depth equal to 60 percent of the ball radius. The raceways shall be filled to capacity with ball bearings.

King bolt. The king bolt shall be of steel and shall have a 1/2-inch minimum diameter. It shall be permanently fixed to the mounting plate by cold heading or upsetting. The king bolt shall not protrude above the mounting plate. Threaded nut and bolt assemblies shall not be acceptable.

Horn assembly. The minimum wall thickness of the steel horn shall be 0.095 inch. The horn shall engage the axle and spanner bushing to prevent rotation.

Lubrication. The swivel assembly and when specified, the axle shall be prelubricated with water repellet grease having a -23° C to 71° C (-10° F to 160° F) operating range.

Lubrication fittings. When specified, the swivel assembly and axle or wheel hub shall have pressure lubrication fittings.

Static load test. The caster shall withstand four times its rated capacity for 1-minute without damage.

Floor marking test. The caster, loaded to four times its rated capacity shall be cycled through six wheel revolutions in each direction. The smooth hardwood or concrete test surface shall show no evidence of treadmarks.

Workmanship. The caster shall be free from imperfections which may impair durability or serviceability.

The issue of ASTM D 2240 in effect on the date of the solicitation shall be used in compliance with these requirements.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 95-580 to the maximum extent practicable.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes: The following options are available and should be specified in contract or ordering documents:

(a) Size and height (dimension D of Figure I) required.
(b) If roller bearing is required.
(c) If steel thread guards are required.
(d) If axle is to be prelubricated.
(e) If lubrication fittings are required.

This description does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this commercial item description, or those which are commercially available, but is intended to cover the types normally procured to meet Federal requirements.

ASTM Standard D 2240 is available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

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MILITARY INTERESTS:

Coordinating Activity

Army - AR

Custodians

Army - AR
Navy - YD
Air Force - 99

Review Activities

DSC - IS
Army - ME
Navy - MC

User Activities

Navy - YD, MC

CIVIL AGENCY COORDINATING ACTIVITIES:

NASA - JFK
VA - VOC

PREPARING ACTIVITY:

GSA - FSS

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